

ALEX SALAMATI

310-469-3268 | Alexsal621836@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Loyola Marymount University, Los Angeles, CA

Expected May 2027

B.S. Computer Engineering, Minor: Computer Science

Relevant Coursework: Computer Architecture (SystemVerilog), Digital Systems, Microprocessors, Circuits I & II, Electronics, Data Structures & Algorithms

Fall 2026: Embedded Systems (Zynq SoC, AXI, HLS), Graduate Machine Learning, Senior Capstone

TECHNICAL SKILLS

Hardware: Verilog, SystemVerilog, RTL design, FSM design, self-checking testbenches, systolic arrays, quantization

Tools: Xilinx Vivado (RTL simulation, waveform debug), Multisim, MPLAB X, Git/GitHub, Linux, oscilloscopes, logic analyzers

Languages: C, C++, Python, MATLAB, Bash, MIPS Assembly

HARDWARE PROJECTS

Folded, Quantized Systolic MAC Array Accelerator | Senior Capstone | Verilog | OpenLane **Sep 2026 – Present**

- Own RTL and architecture on a 3-person team building a quantized systolic multiply-accumulate (MAC) array, the core operation of GPU tensor cores; designing the processing-element datapath (INT8, with an INT4 variant) and the folding-controller FSM that time-multiplexes a small physical array to emulate a larger logical one.
- Planned milestones: verification against a Python/NumPy golden model (Nov to Jan), Xilinx FPGA bring-up (Jan to Feb), then OpenLane RTL-to-GDSII with DRC, LVS, and timing signoff on SkyWater sky130 (Feb to Mar).

MIPS Single-Cycle Processor | SystemVerilog | Vivado | Zynq-7020

Mar 2026 – May 2026

- Designed a 32-bit single-cycle MIPS processor in SystemVerilog (ALU, ALU control, control unit, register file, instruction and data memory, full datapath) supporting a 14-instruction subset: nine R-type ALU operations plus addi, lw, sw, beq, and j, targeting a Zynq-7020.
- Verified in Vivado simulation with self-checking testbenches: 9 assertions over two complete programs (arithmetic, slt, taken and untaken branches, jump) and (load, store, reload), 14 further module-level checks on the register file and instruction memory, and 100 randomized ALU vectors across 10 opcodes reviewed in waveform.

Also: two-stage CMOS op-amp (Multisim, Miller compensation, >60 dB gain, >45° phase margin); ball-balancing robot (PID, OpenCV); SO-ARM101 robot arm (LeRobot).

RESEARCH EXPERIENCE

Undergraduate Researcher, CubeSat Communications | LMU | Advisor: Dr. Vejarano

May 2026 – Jul 2026

- Found and fixed the concurrency bug behind recurring system freezes: a binary semaphore released between 16-byte I2C transfer chunks let concurrent RTOS tasks corrupt messages mid-transfer. Reworked arbitration to hold the bus for the full message; no freeze recurred in any later FlatSat run.
- Owned the communications pipeline (Raspberry Pi payload, dsPIC33 onboard computer, EnduroSat UHF transceiver) in C on Salvo RTOS, building on a prior student's firmware base; verified end-to-end downlink of three telemetry types via AX.25 at 435 MHz, 9.6 kbaud FSK, to a GNU Radio ground station on a USRP N2920.
- Validated the two-way command path: a ground-station uplink command reaches the onboard computer, which polls the payload over I2C and downlinks the jamming report on demand.
- Isolated three independent faults by single-variable testing (shared-buffer race, missing ESTTC carriage return, blocking GPS serial read); ran a byte-survival study establishing the 97-byte register ceiling and mandatory hex-encoding for image data; wrote the 20-page handoff manual.

EXPERIENCE

IT Support Technician | LMU ITS Classroom Support | Work Study

Jun 2024 – Present

- Diagnose and resolve AV, OS, and hardware failures across 20+ smart classrooms, usually in under 5 minutes, while lectures are running; document recurring failure patterns and escalate firmware-level issues to vendors.

LEADERSHIP

Vice President, Additive Manufacturing Club | LMU

2024 – Present

- Lead 3D printing projects (CAD review, slicing, FDM and resin calibration); onboarded and trained 10+ new members.

Member, ROBO Robotics Club and IEEE Student Chapter | LMU

2023 – Present